Lecture 31.08: Choosing K Code — Elbow Method Aur Silhouette Score Implementation [Hinglish]











📌 Topic Overview

Practical implementation of K-selection techniques in Python. Demonstrates looping over K values, computing inertia_ and silhouette_score, automated knee detection with kneed.KneeLocator, and comparative evaluation.

📋 Is Topic Mein Hum Kya Seekhenge?

- $K = 1$ se 20 tak loop chalakar `kmeans.inertia_` collect karna.
- Elbow curve plot karke visually inflection point identify karna.
- `kneed.KneeLocator` library se programmatically exact elbow nikalna (`knee.elbow`).
- $K = 2$ se 20 tak `silhouette_score(X, labels)` calculate karna.
- Silhouette curve plot karke maximum score par optimal K confirm karna.

🔍 Har Ek Point Ka Simple & Detailed Explanation

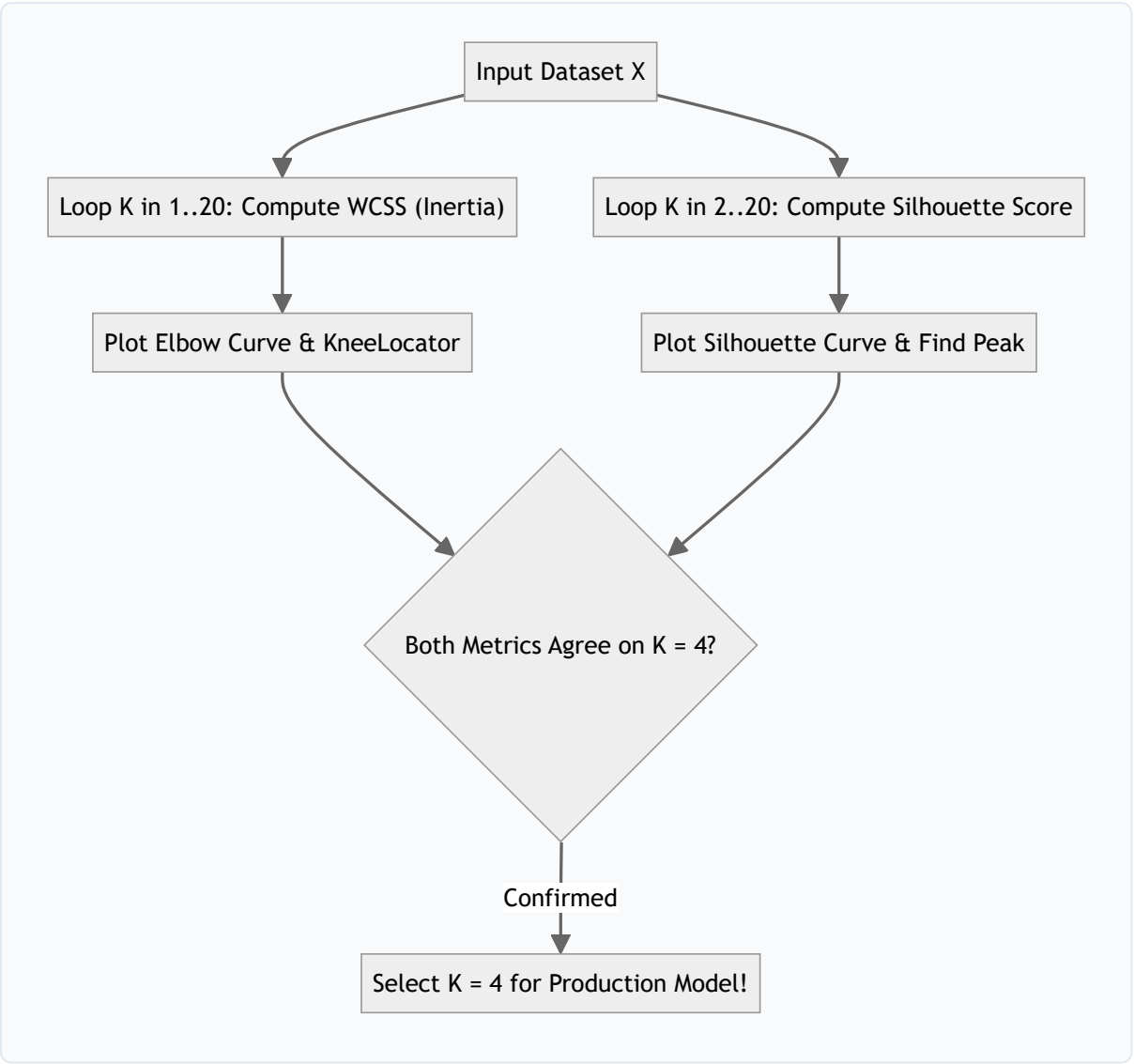
- **Elbow Method Code**: `wcss = []` list banayi gayi aur loop mein $K = 1$ se 20 tak `kmeans = KMeans(n_clusters=k).fit(X)` chalakar `wcss.append(kmeans.inertia_)` store kiya gaya.
- **Elbow Plotting**: `sns.lineplot(x=range(1, 21), y=wcss, marker='o')` se clean elbow graph draw hua.
- **Automated KneeLocator**: Manual aankhon se dekhne ke bajay python library `kneed` use ki gayi: `knee = KneeLocator(range(1, 21), wcss, curve='convex', direction='decreasing')` , jisse exact `knee.elbow = 4` print hua!

- **Silhouette Score Code**: `from sklearn.metrics import silhouette_score` import kiya gaya. $K = 2$ se 20 tak `ss.append(silhouette_score(X, kmeans.labels_))` calculate kiya gaya.
- **Peak Detection**: Silhouette plot mein $K = 4$ par highest peak (~0.79) mili, jo Elbow point ke sath 100% match karta hai!
- **Code Lecture**: Is lecture ka complete runnable notebook lecture_31_08.ipynb mein available hai.

📋 Elbow vs Silhouette Score Code Execution Comparison

Step / Aspect	Elbow Method Implementation	Silhouette Score Implementation
Import	Built-in `kmeans.inertia_`	`from sklearn.metrics import silhouette_score`
Valid K Range	$K \in [1, N]$ (can evaluate $K = 1$)	$K \geq 2$ (minimum 2 clusters required!)
Selection Criterion	Point of maximum curvature (Elbow)	Absolute highest peak value (Global Maximum)
Automation Tool	`kneed.KneeLocator`	`numpy.argmax(scores) + 2`
Output Metric	Inertia (unbounded squared distances)	Normalized coefficient $\in [-1.0, +1.0]$

📊 Dual Evaluation Pipeline for Optimal K



🔢 Mathematical Derivation

Inertia attribute computed by scikit-learn:

$$\text{inertia_} = \sum_{i=1}^{N} \min_{k \in \{1,\ldots,K\}} \|\mathbf{x}_i - C_k\|^2$$

Mean silhouette score across all observations:

$$S_K = \frac{1}{N} \sum_{i=1}^{N} s(i)$$

💻 Standalone Python Code & Verification

Iss module ka complete runnable code Jupyter Notebook mein available hai.